

Public vs. Private Networks — Guided Notes

Unit 1.4.1 | CompTIA Network+ N10-009 Obj. 1.4 | N10-008 1.4

Name: _____ Date: _____ Period: _____

Complete each question using the lesson reading. Write in the blank or answer in the space provided.

Question 1

List the three RFC 1918 private IP ranges and their CIDR notation:

1. _____ (_____)
 2. _____ (_____)
 3. _____ (_____)
-
-
-

Question 2

Private IP addresses are never _____ on the public internet. The reason private addressing exists is that IPv4 only provides approximately _____ total addresses, which is not enough for every device in the world to have a unique public IP.

Question 3

NAT stands for _____. When a device on a private network sends traffic to the internet, the router replaces the _____ IP with its own _____ IP before forwarding the packet.

Question 4

PAT (Port Address Translation) allows an entire network to share a _____ public IP address. It tracks which internal device made which request by assigning unique _____ numbers to each outbound connection.

Question 5

APIPA assigns addresses in the _____ range when a device cannot reach a _____. An APIPA address means the device can only communicate with _____ on the same segment and cannot reach _____.

Question 6

When a technician sees a 169.254.x.x address on a workstation, list three things they should immediately check:

1. _____
2. _____
3. _____

Question 7

The loopback address is _____. Pinging this address tests whether _____ is functioning on the local machine. Traffic sent to this address never leaves the device and never touches _____.

Question 8 — Real World Application

REAL WORLD: A user calls to say they can't access anything on the network or the internet. You remotely view their screen and see their IP address is 169.254.83.42. Walk through your diagnostic steps in order.

Download the answer key from the class server:

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wget http://[SERVER]/resources/network-engineer/1-4-1-public-vs-private-networks/1-4-1-public-vs-private-networks-answer-key.pdf
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